

WASP-39b

R-2117

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_120509 · created 2026-08-11T00:54:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456056.724 +/- 0.061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.055
Transit depth (Rp/Rs)^2	3.61 +/- 2.09 [%]
Orbital Inclination (inc)	89.28 +/- 2.09
Airmass coefficient 1 (a1)	0.66 +/- 0.27
Airmass coefficient 2 (a2)	0.24 +/- 0.24
Scatter in the residuals of the lightcurve fit is	36.53 %
Best Comparison Star	None
Optimal Aperture	3.95
Optimal Annulus	11.05
Transit Duration (day)	0.085 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.055 vs 0.1457 (deviation 0.64σ)
Duration fit vs published	0.085 d vs 0.12 d (deviation 0.76σ)
Mid-time O–C	36.1 min
Depth SNR	2.8
Residual scatter	36.53 %

Light curve

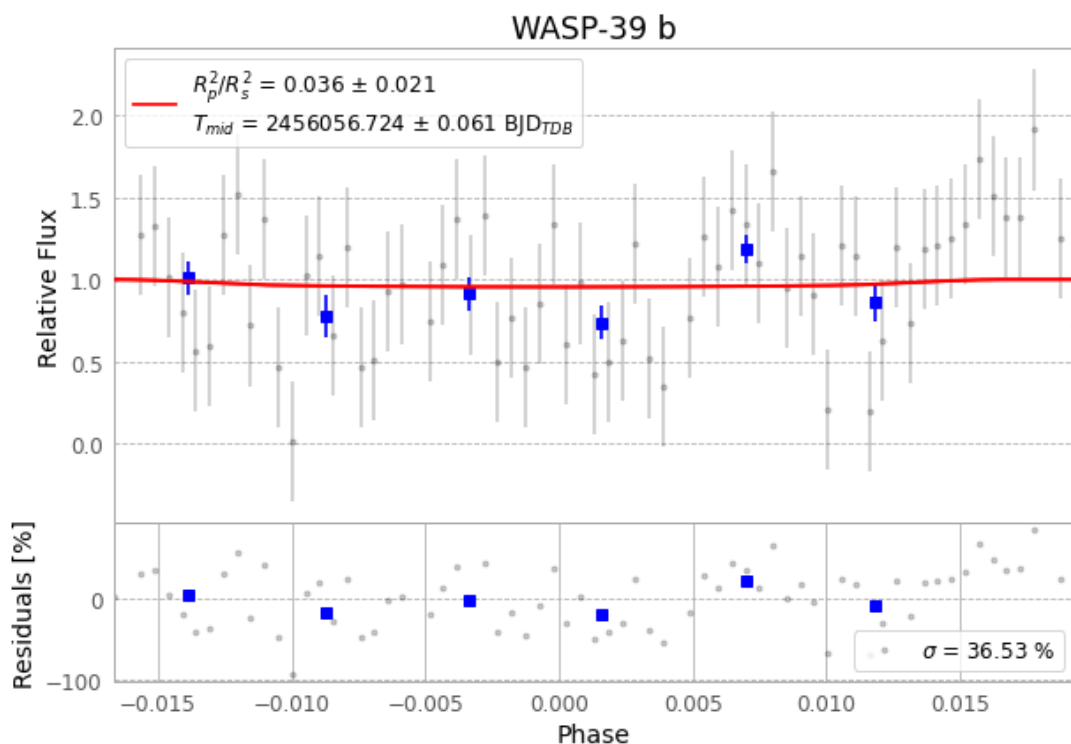


Figure 1 — FinalLightCurve_WASP-39 b_2012-05-08.png (SHA-256 47e6eebd2822f5da...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [263, 369], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e3079d408ad8a552...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

70 FITS frames (46 MB), WASP-39120509033614.FITS → WASP-39120509070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-120509.log (a7c981dd54a0...), FinalLightCurve_WASP-39 b_2012-05-08.png (47e6eebd2822...), FinalLightCurve_WASP-39 b_2012-05-08.csv (6bc787a6c53c...)

Compute resources

Wall time 454.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:54Z.